



Computer Vision Lab — Sampling & Image Quality

Q: A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

1. What is Sampling?

Sampling is the process of converting a **continuous** (real-world) image into a **digital** image by picking pixel values at regular intervals.

 **Continuous Image**
(Real World)



 **Sampling**
(Pick Pixels)



 **Digital Image**
(Pixels)

✓ **High Sampling Rate** → More pixels → Better image quality → More details

✗ **Low Sampling Rate** → Fewer pixels → Loss of details → Blurred image

2. Why Does Blurring Occur?

When the sampling rate is too low:

- Fine image details are **lost**
- Neighboring pixels **merge together**
- Edges become **unclear**
- Small objects **disappear**
- The image looks **blurred or blocky**

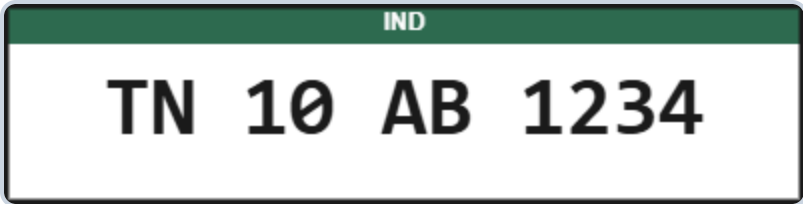
💡 **Key Point:** More pixels capture more details. Better sampling = Better image quality.

3. Example: Vehicle Number Plate

See how different sampling rates affect a number plate image:

Sampling Rate:

100%



400×100

Sharp

99%

TN 10 AB 1234

Resolution

Quality

OCR Accuracy

OCR Result

Sampling Rate	Resolution (W × H)	Quality	OCR Accuracy
Low	50 × 25 pixels	Blurred	45%
Medium	200 × 100 pixels	Average	82%
High	400 × 200 pixels	Sharp	99%

4. How to Correct Blurring?

1

Increase Camera Resolution

Use a higher-resolution camera. Example: Instead of 640×480, use 1920×1080.

2

Increase Sampling Rate

Capture more pixels per unit area. More pixels → Better detail → Clear image.

3

Use Image Interpolation (Software)

Resize using: **Bilinear**, **Bicubic**, or **Lanczos** interpolation. *(Improves appearance but cannot recover lost details.)*

4

Improve Camera Focus

Proper focus ensures the sampled pixels accurately represent the scene.

5. Real-Life Applications Affected by Poor Sampling



**Face
Recognition**

— Face cannot be
identified



**Medical
Imaging**

— Tumors may be
missed



**Self-Driving
Cars**

— Road signs not
detected



**QR
Scanner**

— QR code cannot be
decoded



OCR — Text recognition accuracy drops

6. Python Code — Try It Yourself

Use these OpenCV code snippets in your lab to experiment with sampling and interpolation.



Required Image for Code

Download the sample number plate image to run the code below.



Download Image



How to Run this Code and See the Output

Step 1: Where to run it?

- **Google Colab / Jupyter Notebook (Recommended for Labs):** Create a new notebook. Upload the `number_plate.jpg` file into the Colab files section. Paste the code into a cell and press Shift+Enter. The output images will appear directly below the cell!
- **Local Python Script (.py):** Create a file named `lab_demo.py` in the exact same folder as the downloaded image. Paste the code into the file, open your terminal/command prompt, and run `python lab_demo.py`.

Step 2: Install Libraries (Only if running locally)

```
pip install opencv-python matplotlib numpy
```

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Step 3: How to see the output?

- The code uses `plt.show()` which triggers the output visualization.
- **In Colab/Jupyter:** The image will instantly render inside the webpage.

- **Running Locally:** A new desktop window will pop up showing the result. *Note: You must close the popup window for the script to continue running!*

6.1 Read and Display an Image

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```
import cv2                                # Import OpenCV library for image proces
import matplotlib.pyplot as plt           # Import Matplotlib for displaying image

# Read the image from the disk into a NumPy array
img = cv2.imread('number_plate.jpg')      # cv2.imread reads the image in BGR (Blu
# Convert the BGR image to RGB format since Matplotlib expects RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # This ensures colors look correct when

# Display the image using Matplotlib
plt.imshow(img_rgb)                       # Render the RGB image array
plt.title('Original Image')               # Add a title to the plot window
plt.axis('off')                           # Hide the X and Y axes (ticks and borde
plt.show()                                # Show the actual window on screen

# Print the dimensions of the loaded image
print("Image Shape:", img.shape)          # Outputs (height, width, channels) e.g.
```

6.2 Simulate Low Sampling (Downsample)

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```
# Downsample the image to a tiny 50x25 resolution to simulate a very low sampling rate
low_res = cv2.resize(img, (50, 25), interpolation=cv2.INTER_NEAREST) #
print("Low-Res Shape:", low_res.shape)    # The image is now only 25 pixels high a

# Scale the tiny image back up to 400x200 so we can actually see the blocky pixelation
pixelated = cv2.resize(low_res, (400, 200), interpolation=cv2.INTER_NEAREST) # Upscale

# Create a figure with a specific size (width=10, height=4 inches)
plt.figure(figsize=(10, 4))               # Create a visua

# Create the left subplot (1 row, 2 columns, 1st position)
plt.subplot(1, 2, 1)                       # Position this
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Display the original, high-quality i
```

```
plt.title('Original (High Sampling)') # Set a descript
plt.axis('off') # Hide axis tick

# Create the right subplot (1 row, 2 columns, 2nd position)
plt.subplot(1, 2, 2) # Position this
plt.imshow(cv2.cvtColor(pixelated, cv2.COLOR_BGR2RGB)) # Display the blurred/blocky low
plt.title('Pixelated (Low Sampling)') # Set a descript
plt.axis('off') # Hide axis tick
plt.show() # Render both subplots to the screen
```

6.3 Compare Different Sampling Rates



```
# Define a list of different resolutions to test (Low, Medium, High)
sizes = [(50, 25), (200, 100), (400, 200)] # Width x Height tuples
# Labels for our plot titles
labels = ['Low (50x25)', 'Medium (200x100)', 'High (400x200)'] # Define corresp

plt.figure(figsize=(14, 4)) # Create a wide figure window

# Loop through each size and label simultaneously
for i, (size, label) in enumerate(zip(sizes, labels)): # Iterate throug
    # Step 1: Downsample the original image to the specific 'size' to simulate the sens
    small = cv2.resize(img, size, interpolation=cv2.INTER_AREA)

    # Step 2: Upsample it back to 400x200 so we can compare them all at the same physic
    restored = cv2.resize(small, (400, 200), interpolation=cv2.INTER_NEAREST) # Upsamp

    plt.subplot(1, 3, i + 1) # Create a subplot (1 row, 3 columns, po
    plt.imshow(cv2.cvtColor(restored, cv2.COLOR_BGR2RGB)) # Show the image
    plt.title(label) # Set the title to the current sampling
    plt.axis('off') # Hide the axis

plt.suptitle('Effect of Different Sampling Rates') # Add a main title for the whole win
plt.tight_layout() # Automatically adjust spacing between p
plt.show() # Display the final comparison
```

6.4 Correction Using Interpolation Methods



```

# Start with a heavily downsampled (blurred) low-resolution image
low_res = cv2.resize(img, (50, 25), interpolation=cv2.INTER_AREA) # Apply resize d

# Define a dictionary of the 4 mathematical interpolation methods OpenCV provides for u
methods = { # Map names to O
    'Nearest Neighbor': cv2.INTER_NEAREST, # Fast, but creates blocky pixelated edg
    'Bilinear': cv2.INTER_LINEAR, # Uses 4 nearby pixels, creates smooth b
    'Bicubic': cv2.INTER_CUBIC, # Uses 16 nearby pixels, creates sharp a
    'Lanczos': cv2.INTER_LANCZOS4 # Uses math formulas over 8x8 pixels, hi
}

plt.figure(figsize=(14, 4)) # Create a wide figure to fit 4 images

# Loop through our dictionary of methods
for i, (name, method) in enumerate(methods.items()): # Iterate throug
    # Here is the correction! Upscale the low-res image using the current mathematical
    upscaled = cv2.resize(low_res, (400, 200), interpolation=method)

    plt.subplot(1, 4, i + 1) # Place image in a 1x4 grid
    plt.imshow(cv2.cvtColor(upscaled, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB and show
    plt.title(name, fontsize=9) # Label the image with the method name
    plt.axis('off') # Hide axis tick

plt.suptitle('Interpolation Methods Comparison') # Set a master t
plt.tight_layout() # Automatically
plt.show() # Display the corrected images side-by-s

```

6.5 Measure Image Quality (PSNR)



PSNR = Peak Signal-to-Noise Ratio. It mathematically compares a degraded image to the original.

```

original = cv2.resize(img, (400, 200)) # Set our baseline ground truth image

Test 3 different sampling sizes
for label, size in zip(['Low', 'Medium', 'High'], [(50,25), (200,100), (400,200)]): # Iter
    small = cv2.resize(img, size, interpolation=cv2.INTER_AREA) # Simulate samp
    restored = cv2.resize(small, (400, 200), interpolation=cv2.INTER_CUBIC) # Correct using

# Calculate the mathematical quality score between the original and the restored image

```

```
psnr = cv2.PSNR(original, restored) # Compute Peak Signal-to-Noise Ratio (PSNR)

print(f"{label} Sampling -> PSNR: {psnr:.2f} dB") # Print the result (e.g., "Low Sampling -> PSNR: 20.50 dB")
```

💡 **Output Example:** Low → ~20 dB (poor), Medium → ~30 dB (okay), High → ~∞ dB (identical)



Conclusion

Improper sampling captures fewer pixels, causing **blurred images** and loss of important details. Increasing the sampling rate (higher resolution) and using proper image acquisition techniques produces **clearer images**, improving the accuracy of computer vision tasks.

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This page/site is developed and maintained by Dr. T. M. Usha